a
Volvox carteri (tip)
V. africanus/reticuliferus
Astrephomene (tip)
Pleodorina (tip)
heterocystous Nostocales
land plants
Florideophyceae
Fucophycidae
Pezizomycotina
Agaricomycetes
volvocine extracellular matrix
two-component signalling
protein kinases
plant receptors
transcription factors
fungal metabolism
pherophorin-like (DUF3707) 21 10 6 6
matrix metalloprotease (Peptidase_M11) 7 11 2 5
guanylate cyclase 5 15 5 4 5 1 1
response regulator 1 1 24 2 6 5 3
histidine kinase (HisKA) 1 1 17 2 2 4 3
histidine kinase ATPase (HATPase_c) 1 1 1 21 2 2 4 3
GAF domain 16 1 1
pentapeptide repeat 17 2
protein kinase (Pkinase) 15 20 9 10 16 42 8 36 9 13
Tyr/Ser-Thr kinase (PK_Tyr_Ser-Thr) 9 11 5 6 22 1 11 1 2
leucine-rich repeat (LRR_8) 1 3 14 4 13
F-box 2 1 16 3 3 3
C2H2 zinc finger 2 16 2 3 3
bZIP 2 3 9 2 2
Zn2Cys6 (Zn_clus) 7 19 9
MFS transporter 3 1 1 3 5 5 21 27 16
short-chain dehydrogenase 1 1 3 4 2 15 14
cellulose-binding (CBM_1) 2 12 3
volvocine germ–soma
other simple → complex (internal branches)
families gained or duplicated on the event branch that carry the domain (count)
40
35
30
25
20
15
10
5
0
b
Kinases: one new family at each of 5 of 7 events
(recurrence p = 0.023, q = 0.06)
V. africanus/reticuliferus 1 19
heterocystous Nostocales 0 16
land plants 1 41
Florideophyceae 1 7
Fucophycidae 1 35
Pezizomycotina 0 9
Agaricomycetes 1 12
new (sequence-novel)
duplicated
0
10
20
30
40
families with a protein-kinase domain (count)
c
Most new families have no known domain
Volvox carteri (tip) 18 of 1,300
V. africanus/reticuliferus 4 of 104
Astrephomene (tip) 29 of 892
Pleodorina (tip) 167 of 1,621
heterocystous Nostocales 53 of 308
land plants 132 of 259
Florideophyceae 120 of 709
Fucophycidae 63 of 493
Pezizomycotina 160 of 417
Agaricomycetes 28 of 139
0
20
40
60
new families with any Pfam domain (%)